WDRO-Regularized Deep Hedging: Metrics, Algorithms, and Results

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Contents

1 Introduction **1**
- 1.1 Market models 1

2 Metrics **2**

3 Algorithm **3**
- 3.1 Dual objectives 3
- 3.2 ULA proposal mechanism 3

4 Main Results **3**
- 4.1 Toy horizon (n_time=5): which objective, which question 3
- 4.2 Realistic horizons (n_time=20, 30) 4
- 4.3 Is the edge just an 8×-samples confound? 5
- 4.4 ULA importance-sampling ablation (n_time=30) 5
- 4.5 ULA step-size rescue at k=2 (coarse, $n = 3$ seeds, unconfirmed) 6
- 4.6 Summary across investigations 7

5 Conclusion **7**

1 Introduction

We consider discrete-time robust hedging: a strategy δ chooses hedge positions to minimize loss under the *worst* plausible market law in a neighborhood of a reference model $\hat{\pi}$,

$$\inf_{\delta \in M} \sup_{\pi \in A(\hat{\pi}, \varepsilon, \mathcal{V})} \mathbb{E}_{\pi} \left[\mathrm{Loss}^{\mathrm{dist}}(\delta) \right], \tag{1}$$

where Loss $= -$PnL and the ambiguity set $A(\hat{\pi}, \varepsilon, \mathcal{V})$ is a Wasserstein/KL ball of radius ε around $\hat{\pi}$. The strategy δ is a small MLP trained on simulated trajectories from two reference models: Black–Scholes (BS) and Heston. The outer supremum is approximated by a family of *dual* objectives (entropic tilting, joint/relaxed CVaR reformulations) evaluated on proposal samples drawn either directly from a reference kernel or via an Unadjusted Langevin Algorithm (ULA) chain. This report consolidates the metrics, the algorithmic components, and the main quantitative findings produced across this investigation.

1.1 Market models

Black–Scholes. The stock follows exact geometric Brownian motion, simulated without discretization error:

$$dS_t = \sigma S_t\, dW_t, \qquad S_{t+\Delta t} = S_t \exp\Big(-\tfrac{1}{2}\sigma^2 \Delta t + \sigma\sqrt{\Delta t}\, Z_t\Big), \quad Z_t \sim \mathcal{N}(0,1). \tag{2}$$

Heston. Stochastic-variance model, with the CIR variance process sampled exactly (noncentral chi-square via a Poisson-mixture-of-gammas representation, not an Euler discretization) and the log-stock update using the standard conditional-on-variance-path decomposition:

$$dV_t = \kappa(\theta - V_t)\, dt + \xi\sqrt{V_t}\, dW_t^V, \tag{3}$$

$$dS_t = \sqrt{V_t}\, S_t \left(\rho\, dW_t^V + \sqrt{1-\rho^2}\, dW_t^{\perp}\right), \tag{4}$$

where κ is mean-reversion speed, θ the long-run variance, ξ the vol-of-variance, and ρ the spot-variance correlation. The stock update uses the piecewise-constant (left-endpoint) integrated-variance approximation $\int_t^{t+\Delta t} V_s\, ds \approx V_t \Delta t$ within each daily step; the variance transition itself is exact.

Table 1: Default `MarketConfig` parameters (both models share spot/strike/horizon).

Parameter	Symbol	Default	Applies to
spot	S_0	100.0	both
strike	K	100.0	both
horizon	T	$n_{\text{time}}/365$ (e.g. 30/365)	both
volatility	σ	0.20	Black–Scholes
mean-reversion	κ	1.0	Heston
long-run variance	θ	0.04	Heston
vol-of-variance	ξ	2.0	Heston
correlation	ρ	-0.7	Heston
initial variance	V_0	0.04	Heston

2 Metrics

Every experiment reports two numbers, both CVaR at level 0.95 (*CVaR-95*) of the hedging loss, i.e. the mean loss in the worst 5% of simulated outcomes:

- **in_distribution**: CVaR-95 evaluated under the same market law used for training (BS: volatility = 0.20; Heston: base parameters).
- **worst_extreme**: the *maximum* CVaR-95 over a grid of out-of-distribution stress scenarios (the strategy is trained once, then evaluated, unmodified, on each stressed market). Lower is better for both metrics.

Each Heston stress scenario overrides all four model parameters at once (not a 1-D grid over a single parameter): mean-reversion κ and vol-of-variance ξ move in the direction that makes the variance process more volatile and slower to revert, long-run variance θ rises (shifting the whole variance level up), and correlation ρ moves further negative (steeper leverage effect, i.e. a bigger stock drop coincides with a bigger variance spike). The four named scenarios are nested in severity: *moderate* $\rightarrow$ *ultra* monotonically worsens every parameter simultaneously.

Table 2: Black–Scholes stress grid used for `worst_extreme` (volatility σ; in_distribution value 0.20 underlined).

$\sigma \in \{0.10,\ \underline{0.20},\ 0.30,\ 0.60,\ 0.90,\ 1.20,\ 1.50\}$

Table 3: Heston stress scenarios used for `worst_extreme` (default row is in_distribution; each stress row overrides all four columns simultaneously).

Scenario	θ (long-run var.)	ξ (vol-of-var.)	ρ (corr.)	κ (mean-rev.)
default (in_distribution)	0.04	2.0	-0.7	1.0
moderate	0.06	2.5	-0.8	0.75
severe	0.08	3.0	-0.9	0.5
extreme	0.15	5.0	-0.99	0.2
ultra	0.25	8.0	-0.999	0.1

3 Algorithm

3.1 Dual objectives

The sup in (1) is approximated per-anchor by sampling n proposals in a neighborhood of each training path (the "anchor") and aggregating their losses with one of the dual objectives in Table 4. Each objective factors into an *inner* aggregation over the n proposals around one anchor, and an *outer* aggregation over the training batch.

Table 4: Dual objectives implemented in `wdro.py`.

Objective	Inner aggregation (per anchor)	Outer aggregation (over batch)
`entropic`	log-sum-exp tilt (worst-of-neighborhood)	plain mean
`joint_cvar`	log-sum-exp tilt on the tail power	CVaR-α (Rockafellar–Uryasev)
`relaxed_kl_cvar`	log-sum-exp tilt (worst-of-neighborhood)	CVaR-α (Rockafellar–Uryasev)
`plain_average`	unweighted mean, no tilt	plain mean
`plain_cvar_of_average`	self-normalized weighted mean, no tilt	CVaR-α
`nominal` (baseline)	— (no perturbation)	CVaR-α

`plain_average` and `plain_cvar_of_average` are confound-control objectives: they keep the same 8× per-step sample budget and perturbation kernel as `entropic`/`relaxed_kl_cvar` but remove the entropic tilt, isolating the effect of "more samples" from the effect of "tilting toward the worst neighbor."

3.2 ULA proposal mechanism

Given an anchor ξ, proposals are drawn either from a single reference kernel ($k = 0$ Langevin steps: $\tilde{\xi} = \xi + \sigma \cdot \text{noise}$, log-weight $= 0$), or via a k-step Langevin chain ($k \geq 1$) targeting the tilted density:

$$\tilde{\xi}_{t+1} = \tilde{\xi}_t + \tfrac{1}{2}\eta\, \nabla_\xi \log \pi_{\text{target}}(\tilde{\xi}_t) + \sqrt{\eta}\, z_t, \qquad z_t \sim \mathcal{N}(0, I). \tag{5}$$

For $k \geq 1$, an importance-sampling (IS) correction is applied,

$$\log w = \log \pi_{\text{reference}}(\tilde{\xi}_k) - \log q_{\text{transition}}(\tilde{\xi}_k), \tag{6}$$

comparing the *last* transition kernel against the *original* reference kernel (not a full accumulated-path density ratio). A configuration flag, `use_importance_sampling` (default `True`), was added this investigation to force $\log w = 0$ regardless of k, isolating whether this (approximate) correction, rather than the extra Langevin step itself, drives $k > 1$'s known underperformance.

4 Main Results

4.1 Toy horizon (n_time=5): which objective, which question

At a 5-day toy horizon, paired 10-seed comparisons (seeds 2300–2309) give the headline trade-off in Table 5.

Table 5: Toy horizon ($n_{\text{time}} = 5$), best-tuned λ per objective/market.

Market	Config	in_distribution	worst_extreme
Heston	nominal	2.18	4.01
Heston	entropic ($\lambda = 0.04$)	2.48 (+14%)	2.94 (−27%)
Heston	relaxed_kl_cvar ($\lambda = 0.02$)	**2.00** (−8%)	**2.69** (−33%)
BS	nominal	1.79	51.1
BS	entropic ($\lambda = 0.1$)	2.62 (+46%)	**30.2** (−41%)
BS	relaxed_kl_cvar ($\lambda = 0.1$)	**1.75** (−2%)	45.8 (−10%)

`relaxed_kl_cvar`'s in-distribution advantage over `entropic` is large and significant (Heston: $p = 1.5 \times 10^{-8}$); its worst-case tail protection ties `entropic`'s on Heston ($p = 0.16$–0.19) but is clearly weaker on BS. **Recommendation at this horizon**: `entropic` if worst-case risk is the only criterion; `relaxed_kl_cvar` if in-distribution performance also matters (especially on Heston, tail protection at near-zero nominal cost).

4.2 Realistic horizons (n_time=20, 30)

The above was re-tested at $n_{\text{time}} = 20$ and 30 (paired 10-seed, seeds 4200–4309; Table 6).

`nominal_bigbatch` (batch_size=512, no perturbation) isolates "more real i.i.d. samples per step alone," mirroring §4.3's toy-horizon comparison. Unlike at $n_{\text{time}} = 5$ (where the effect was modest, in-distribution-only, ≤5%), at the realistic horizon the batch-size effect is substantially larger on in-distribution loss in both markets (−12% to −23%, $p \leq 0.006$ throughout) and now reaches significance on `worst_extreme` for Heston ($p = 0.004$–0.049), though not for BS (a tie in both directions). `relaxed_kl_cvar (k=0)` still beats `nominal_bigbatch` on worst_extreme in every cell, so the WDRO mechanism's tail-risk benefit is not simply subsumed by a bigger batch at this horizon — consistent with §4.3's toy-horizon finding that sample count alone does not explain the WDRO objectives' full advantage.

Key result 1: the toy-horizon finding (§4.1) replicates cleanly at every horizon tested (5, 20, 30 days) — `relaxed_kl_cvar` wins in-distribution ($p < 0.01$ throughout) and ties `entropic` on worst-case tail protection. **Key result 2 (new)**: BS `entropic`'s tail-risk edge over nominal, present at $n_{\text{time}} = 5$ (30.2 vs. 51.1), *disappears* at $n_{\text{time}} = 20/30$ (worst_extreme ties nominal, $p = 1.00$, 0.70)

Table 6: Realistic horizons, paired 10-seed (seeds 4200–4309; `nominal_bigbatch` uses the same seeds, paired directly against `nominal`).

Market	Config	in_distribution	worst_extreme
BS, $n_{\text{time}} = 20$	nominal	6.82	341
BS, $n_{\text{time}} = 20$	nominal_bigbatch (batch=512)	5.28 (−23%, $p = 0.0020$)	379 (+11%, tie)
BS, $n_{\text{time}} = 20$	entropic	10.76 (+58%, worse)	419 (+23%, tie)
BS, $n_{\text{time}} = 20$	relaxed_kl_cvar (k=0)	**5.24** (−23%)	**226** (−34%)
BS, $n_{\text{time}} = 30$	nominal	10.23	625
BS, $n_{\text{time}} = 30$	nominal_bigbatch (batch=512)	**8.41** (−18%, $p = 0.0020$)	570 (−9%, tie)
BS, $n_{\text{time}} = 30$	entropic	14.94 (+46%, worse)	569 (−9%, tie)
BS, $n_{\text{time}} = 30$	relaxed_kl_cvar (k=0)	8.55 (−16%)	**529** (−15%, $p = 0.16$)
Heston, $n_{\text{time}} = 20$	nominal	6.90	27.3
Heston, $n_{\text{time}} = 20$	nominal_bigbatch (batch=512)	**5.51** (−20%, $p = 0.0039$)	16.0 (−42%, $p = 0.0039$)
Heston, $n_{\text{time}} = 20$	entropic	6.82 (−1%, flat)	13.5 (−51%)
Heston, $n_{\text{time}} = 20$	relaxed_kl_cvar (k=0)	5.87 (−15%)	**11.0** (−60%)
Heston, $n_{\text{time}} = 30$	nominal	10.01	54.2
Heston, $n_{\text{time}} = 30$	nominal_bigbatch (batch=512)	8.86 (−12%, $p = 0.0059$)	41.6 (−23%, $p = 0.049$)
Heston, $n_{\text{time}} = 30$	entropic	9.75 (−3%, flat)	31.0 (−43%)
Heston, $n_{\text{time}} = 30$	relaxed_kl_cvar (k=0)	**7.75** (−23%)	**20.7** (−62%)

while its in-distribution cost worsens ($p = 0.002$ both horizons) — confirmed not a lambda-tuning artifact (grid searched 0.1×–8× around the toy optimum, plus an extension down to 0.00125×, no improvement found). Heston's `entropic` tail-risk edge over nominal, by contrast, holds at both longer horizons ($p = 0.037$–0.002).

4.3 Is the edge just an 8×-samples confound?

`relaxed_kl_cvar`/`entropic` both forward-pass each anchor 8 times per step (one per proposal) vs. `nominal`'s 1 pass. Two confound-control objectives were added: `plain_average` (8× samples, no tilt, plain-mean outer aggregation, matching `entropic`'s structure) and `plain_cvar_of_average` (8× samples, no tilt, CVaR-α outer aggregation, matching `relaxed_kl_cvar`'s structure exactly — added after the first control was found to differ from `relaxed_kl_cvar` in *two* variables at once: sample count *and* outer-aggregation type).

Table 7: Toy horizon, paired 10-seed (seeds 6000–6009), mean±std.

Config	BS in_dist	BS worst	Heston in_dist	Heston worst
nominal	1.789 ± 0.023	48.86 ± 5.72	2.142 ± 0.044	4.04 ± 1.14
nominal_bigbatch (batch=512)	1.768 ± 0.008	48.73 ± 3.65	2.032 ± 0.025	3.11 ± 0.51
plain_average (8×, no tilt)	21.51 ± 5.80	982 ± 342	19.46 ± 8.72	27.7 ± 26.1
entropic	3.195 ± 0.377	44.48 ± 9.36	2.584 ± 0.227	3.32 ± 0.61
plain_cvar_of_average (8×, no tilt)	1.747 ± 0.018	44.67 ± 5.81	2.068 ± 0.047	3.184 ± 0.506
relaxed_kl_cvar	1.751 ± 0.017	40.64 ± 3.88	2.000 ± 0.031	2.76 ± 0.68

The single-variable comparison — `relaxed_kl_cvar` vs. `plain_cvar_of_average`, holding both sample count and outer-aggregation type fixed, varying only the inner tilt — is decisive in 3 of 4 market/metric cells: BS worst_extreme ($p = 0.0137$, relaxed wins), Heston in_distribution

($p = 0.0059$, relaxed wins), BS in_distribution (tie, $p = 0.43$, expected — the tilt targets tail regions, not the average case); the 4th cell (Heston worst_extreme, $p = 0.084$, 80% win rate) reads as underpowered-in-the-same-direction rather than a genuine null. **Verdict: the "it's just** $8\times$ **samples" hypothesis is not supported** — the entropic tilt does real, distinct work beyond raw sample count, under both outer-aggregation conventions tested (comparison against `plain_average`, $p \leq 0.024$ everywhere, and against `plain_cvar_of_average` above). Separately, `nominal_bigbatch` vs. `nominal` shows raw sample count alone gives a real but modest in-distribution benefit (Heston $\sim$5%, $p = 0.0020$; BS $\sim$1%, $p = 0.027$), with no worst-case benefit at either market.

4.4 ULA importance-sampling ablation (n_time=30)

Does the (approximate) IS correction, rather than the extra Langevin step, explain why $k = 2$ underperforms $k = 1$ (an established finding at the toy horizon)? Tested `relaxed_kl_cvar` at $n_{\text{time}} = 30$, $k \in \{1, 2\}\times$ `use_importance_sampling`$\in \{\text{True}, \text{False}\}$, paired 10-seed (seeds 7000–7009), reusing established λ (BS: 0.025, Heston: 0.002).

Table 8: Means, $n = 10$ paired seeds.

Config	BS in_dist	BS worst	Heston in_dist	Heston worst
nominal	9.88	497	9.53	67.3
k1_is_on (default)	9.21	419	8.83	24.7
k1_is_off	9.29	449	8.56	23.8
k2_is_on	9.74	615	9.64	33.5
k2_is_off	9.68	372	10.24	37.5

Toggle is a no-op at $k = 1$ ($k1_{\text{is_off}}$ vs. $k1_{\text{is_on}}$: $p \geq 0.12$ everywhere). **Toggling IS off at** $k = 2$ **does not recover** $k = 1$**-level performance**: on Heston (the market with the cleanest $k1$-vs-$k2$ gap), disabling IS makes in-distribution CVaR significantly *worse* ($p = 0.0003$); on BS the effect is a noisy, borderline improvement on worst_extreme only ($p = 0.054$ ttest / 0.014 Wilcoxon, driven by 2 of 10 seeds), still not beating $k = 1$. **Verdict: the IS-correction-bias hypothesis is not supported** as the primary driver of $k > 1$'s underperformance.

4.5 ULA step-size rescue at k=2 (coarse, $n = 3$ seeds, unconfirmed)

A follow-up coarse sweep (3 seeds, no formal significance testing) at $k = 2$, $n_{\text{time}} = 30$, varying `ula_step_size` $\in \{0.02, 0.01, 0.005, 0.0025, 0.00125\}$ (default 0.02) found a striking, near-monotonic improvement as the step size shrinks (Table 9), recovering to — and in several cells beating — the $k = 1$ baseline of §4.4.

Table 9: $k = 2$, `use_importance_sampling`=True, means over 3 seeds.

`ula_step_size`	BS in_dist	BS worst	Heston in_dist	Heston worst
0.02 (default)	9.41	352	9.65	23.3
0.01	8.60	281	8.71	19.0
0.005	8.51	449	7.63	17.8
0.0025	8.51	654	7.90	16.4
0.00125	8.18	386	7.45	18.0

For reference, $k = 1$ default-step-size means were 9.21/419 (BS) and 8.83/24.7 (Heston). The smallest step sizes tested *beat* $k = 1$ on in-distribution loss in both markets, and are competitive on worst_extreme (BS's worst_extreme is noisy at $n = 3$, no clean monotonic trend). A matched `use_importance_sampling`=False sweep at the same grid showed **no consistent IS on/off difference at any step size** (sign flips across cells) — reinforcing §4.4's conclusion that step size, not the IS toggle, is the operative lever. This result is **not yet paired-confirmed with more seeds** and is reported here as a promising, preliminary lead.

To check whether the step-size effect is specific to $k = 2$ or a general lever, the identical sweep (same grid, same 3 seeds, `use_importance_sampling`=True) was run at $k = 1$ (Table 10). Nominal is shown for reference. **The step-size effect is not specific to** $k = 2$: $k = 1$ shows the same near-monotonic improvement as step size shrinks, reaching a similar or better optimum (BS in_dist 7.41 at 0.00125, beating $k = 2$'s best of 8.18; Heston in_dist 7.63 at 0.005, matching $k = 2$'s best of 7.45 within sweep noise). This suggests smaller `ula_step_size` is beneficial regardless of k, and the earlier $k = 2$-vs-$k = 1$ comparison (§4.4) at the shared default 0.02 conflated two effects (chain length and step size) rather than isolating chain length alone.

Table 10: $k = 1$ vs. $k = 2$ step-size sweep, `use_importance_sampling`=True, means over 3 seeds (`nominal` for reference, no step size).

`ula_step_size`	k	BS in_dist	BS worst	Heston in_dist	Heston worst
–	nominal	9.88	497	9.53	67.3
0.02 (default)	1	9.47	332	8.94	23.0
0.01	1	8.96	**265**	7.85	18.4
0.005	1	8.13	429	7.77	**16.4**
0.0025	1	9.00	424	7.88	18.1
0.00125	1	**7.41**	411	7.63	17.4
0.02 (default)	2	9.41	352	9.65	23.3
0.01	2	8.60	281	8.71	19.0
0.005	2	8.51	449	7.63	17.8
0.0025	2	8.51	654	7.90	**16.4**
0.00125	2	8.18	386	**7.45**	18.0

4.6 Summary across investigations

Table 11: Headline worst_extreme numbers across horizons/investigations (best relaxed_kl_cvar configuration each time).

Investigation	n_{time}	BS worst_extreme	Heston worst_extreme
Toy horizon (§4.1)	5	45.8 (nominal: 51.1)	2.69 (nominal: 4.01)
Realistic horizon (§4.2)	30	529 (nominal: 625)	20.7 (nominal: 54.2)
Confound isolation (§4.3)	5	40.64 (nominal: 48.86)	2.76 (nominal: 4.04)
IS ablation, $k = 1$ (§4.4)	30	419 (nominal: 497)	24.7 (nominal: 67.3)
Step-size sweep, $k = 2$, best (§4.5)	30	281	16.4

5 Conclusion

Current production recommendation: `relaxed_kl_cvar` (alpha=0.9, kl_epsilon=0.03, cost_power=1, gaussian proposal, $k = 1$ ULA step), λ tuned per market (Heston: 0.02 at $n_{\text{time}} = 5$, 0.002 at $n_{\text{time}} = 30$; BS: 0.1 at $n_{\text{time}} = 5$, 0.025 at $n_{\text{time}} = 30$). This is the one configuration with reviewer-verified, paired-significant evidence of beating nominal on both in-distribution and worst-case axes, across every horizon tested (5, 20, 30 days), and confirmed (§4.3) to be more than a sample-count artifact.

Open threads:

- BS `entropic`'s worst-case tail-risk edge, part of the toy-horizon guidance, does not hold at realistic horizons ($n_{\text{time}} = 20/30$) — no confirmed objective yet delivers BS tail-risk-only hedging at these horizons.
- The $k = 2$ ULA step-size rescue (§4.5) is a promising but unconfirmed lead (coarse, $n = 3$ seeds): a properly tuned $k = 2$ may match or beat $k = 1$, contrary to prior guidance that fixed $k = 1$ as the default. Needs a paired multi-seed confirmation before revising that guidance.
- Heston's apparent `relaxed_kl_cvar`-over-`nominal` in-distribution advantage in the original toy-horizon study is partly, but not entirely, attributable to sample count rather than the WDRO mechanism itself (§4.3); the tail-risk advantage there is more tilt-specific.